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Why do birds use green nest material? A systematic review and meta-analysis of experiments

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Why do birds use green nest material? A systematic review and meta-analysis of experiments

Abstract

Many animals build nests. As external structures that can influence survival and reproduction, nests are often considered extended phenotypes. Birds are key examples of nest builders, and some species add green plant material to their nests. Yet, the adaptive value of this behaviour remains debated. Non-mutually exclusive hypotheses propose roles in courtship signalling, parasite defence, and enhancement of offspring condition through pharmacological effects independent of parasite reduction. Here, we conducted a pre-registered systematic review and meta-analysis of 28 experimental studies (26 published, 2 unpublished), spanning seven bird species and 274 effect sizes, to test whether green nest material enhances fitness and to evaluate competing functional explanations. Our meta-analysis shows that green nest material can increase fitness; however, this effect varied depending on the fitness proxy investigated, being strongest for morphological proxies. We found no compelling evidence to preferentially support the courtship, nest protection, or drug hypothesis. Nonetheless, experimental design (i.e., treatment–control comparison type) was the moderator explaining most effect size variation, challenging the traditionally held role of aromatic compounds in the fitness benefits of green nest material. Our synthesis provides evidence for the adaptive significance of green nest material and highlights the need for further research into the underlying mechanisms.

Introduction

Nest-building is fundamental across a diverse range of animals, including arthropods (e.g., ants, spiders), mammals (e.g., platypuses, rodents), and perhaps more famously, birds. Nests provide structural support for eggs, parents, and offspring, and protect them against predators and other environmental pressures, and are thus key to reproductive success and survival [1–3]. Most birds use organic plant materials such as twigs, branches, grass, and moss to build their nests. Some even incorporate inorganic (e.g., mud, stones), animal-derived (e.g., feathers, snake skins), and anthropogenic materials (e.g., plastic, cigarette butts) [4–7]. These additions are considered non-structural. Beyond their role as physical shelters, nests are complex tools shaped by both natural and sexual selection for multiple functions. Indeed, nests can play a key role in sexual display, and features such as nest size have been shown to act as sexually selected signals of parental quality (e.g., [8,9]). Nest characteristics may even convey information interspecifically, for example, when brood parasites use host nest traits as cues to host quality [10]. Nest-building behaviour is increasingly recognised as an expression of the extended phenotype (reviewed by [11]), and phylogeny has been shown to be an important factor explaining nest morphology [12]. Yet, despite hundreds of scientific articles on bird nests and nesting behaviour being published each year, much remains unknown [13].

An intriguing nesting behaviour observed in some bird species is the addition of fresh, often aromatic, plant material to the nest. This green nest material (*hereafter GNM; also called green plant material or greenery*) often includes leaves, sprigs, and herbs. It is typically added from the start of nest building until egg-laying, although some species like the blue tit (*Cyanistes caeruleus*) keep adding it afterwards. Unlike the structural nest materials such as twigs and mud, GNM is selected from a small, non-random subset of local flora [14–16]. The selected plants often contain high levels of aromatic volatile compounds (commonly used by songbirds, [17]) or secondary metabolites such as oils and resins (commonly used by raptors, [18]). This suggests that GNM is unlikely to be gathered opportunistically, and collecting and replenishing these uncommon plants, sometimes over weeks, requires extra time and, in some cases, flying outside the territory [19]. Thus, the general costs associated with nest-building [20], and the additional effort, time, and energy devoted to adding these non-structural materials suggest an adaptive function of GNM.

Multiple hypotheses have been proposed to explain the adaptive function of GNM (reviewed in [21,22]). First, the courtship hypothesis suggests that males use GNM as a visual or olfactory signal to attract females, maintain established pairs [23], or advertise nest-site occupancy to other males (e.g., [24–26]), with olfactory signalling being particularly relevant in cavity-nesting species [27]. For instance, a seminal study by [28] showed that male European starlings (*Sturnus vulgaris*) carry fresh plant material to their nests while singing and displaying to females. Subsequent studies in European and spotless starlings (*Sturnus unicolor*) supported this view by showing that females often prefer nests with fresh GNM. Unpaired males conspicuously bring GNM to nests during courtship [26] and add more GNM when potential mates are nearby [24]. Altogether, these studies offered support for the courtship hypothesis, which was later extended to account for the observed GNM addition solely by females in blue tits [29].

Second, the most widely tested hypothesis is the nest protection hypothesis, which proposes that the volatile compounds in GNM reduce parasitic and pathogenic loads in nests, enhancing offspring survival. This hypothesis was first suggested by [30] in a comparative study in raptors (Falconiformes). Soon after, experimental studies of European starlings showed a preference for GNM containing insect-repellent and antimicrobial volatiles, and [14] proposed three key criteria for evaluating the nest protection hypothesis: selective plant use, bioactive chemical properties, and demonstrable adaptive benefits. Observational and experimental studies in raptors [18,31] and passerines [17,32–34] have shown mixed (i.e., contradictory) results. The drug hypothesis was later proposed because GNM had been shown, in some cases, to improve offspring condition without reducing parasite abundance [35,36]. This alternative hypothesis suggests that GNM volatiles directly enhance offspring immune function and/or health independently of parasite or pathogen control [37]. While both the nest protection and drug hypotheses suggest that GNM enhances offspring survival, the drug hypothesis attributes this effect to a pharmacological mechanism, whereas the nest protection hypothesis emphasises parasite and pathogen repellency. Importantly, these two hypotheses are not mutually exclusive, and their potential effects on offspring survival are difficult to disentangle because improved nestling condition may result either directly from pharmacological action or indirectly through reduced parasite or pathogen load. Therefore, in our meta-analysis, we combined them under the “parental care hypothesis”.

Other less explored hypotheses include a closely related female protection hypothesis proposed in blue tits [38,39], suggesting that plant volatiles may primarily benefit the incubating female by reducing her parasite exposure and infection risk [40]. Several hypotheses have also been proposed that do not invoke the aromatic properties of

GNM, including enhancement of nest crypticity [41], thermal insulation or sun protection for eggs and offspring [42], and a structural function for the added material [43]. However, these alternative hypotheses have received comparatively little empirical attention and therefore fall outside the scope of the present meta-analysis.

Despite decades of research, abundant observational and experimental studies, and two narrative reviews [21,22], the overall adaptive significance of GNM, its generality across species, and the relative importance of each of the suggested mechanisms are still unknown. To address this gap, we conducted a pre-registered systematic review and meta-analysis of experiments testing the effect of GNM on various fitness proxies across birds. Our meta-analysis explicitly evaluates the overall evidence for the adaptive significance of GNM, and assesses the relative importance of the main suggested hypotheses: the courtship hypothesis and parental care hypothesis. In addition, we tested potential moderators predicted to explain the observed context-dependency (i.e., high heterogeneity among effect sizes). Specifically, we tested whether the effect of GNM varied depending on the experimental design used, the timing of GNM addition, the focal species tested, the category of fitness proxy investigated (e.g., morphological, reproductive, etc.), and the type of parasite studied. Our meta-analysis of experimental studies is the first quantitative synthesis testing the causal relationship between GNM and fitness across birds that show this behaviour.

Methods

Registration and reporting

We pre-registered our study protocol on the Open Science Framework [44] before exploring or analysing the data. The pre-registration details our objectives, hypotheses, predictions, methods, and planned analyses. Unless stated otherwise, we adhered to the registered protocol. We followed the PRISMA-EcoEvo guidelines [45] for reporting this study (Table S1; Figure S1).

Information source and search

We conducted a systematic search on Web of Science Core Collection and Scopus on the 7th of September, 2022 and updated it on the 12th of August, 2024. The keyword search query was designed as a three-block query including terms related to our treatment of interest ((“green*” OR “herb*” OR “aromatic*”) AND “nest*”), population of interest (“bird*” OR “aves” OR “avian” OR “ornithol*” OR “passerine*” OR “passeriform*” OR “songbird*” OR list of all bird genera) and to limit the search results to experimental studies only (“experiment*” OR “manipulat*”) (see complete search string in Figure S15). Although our search terms were in English, we did not restrict eligibility by language. We planned to include studies in English, Spanish, Portuguese or Hindi if they were retrieved, but all studies identified were in English. We searched through titles, abstracts and keywords without restrictions on publication date or study type. We validated our search using an initial library of 15 relevant articles identified before conducting the searches (Figure S2), ensuring all were retrieved. Additionally, we used the R package ‘litsearchr’ v1.0.0 [46], which confirmed our search queries. After removing 165 duplicates using the R package 'revtools' v0.4.1 [47], we identified 436 unique records.

Eligibility criteria

All titles and abstracts were equally allocated and screened independently by at least two of the three screeners (SD, TR, JMGS) using 'revtools' following a pre-designed decision tree with all inclusion/exclusion criteria (Figure S3). We included all studies on birds that experimentally manipulated the GNM (both aromatics and non-aromatic). Forty articles passed the title-and-abstract screening and were further evaluated during the full-text screening, where each article was independently screened by a single screener, with another screener verifying the procedure. We used the same criteria as above (Figure S3) with an additional evaluation of whether the variable of interest studied in the article was in our list of fitness proxies of interest (Table S2). All conflicts in screening decisions (1% for title-and-abstract screening, and none for full-text screening) were discussed and resolved. Twenty-three articles passed the full-text screening, and we identified three additional eligible articles recommended by two colleagues, likely missed because their experimental nature was not indexed in the searchable metadata. In all, 26 published articles were eligible for our meta-analysis. We also obtained two additional unpublished studies, resulting in a total of 28 studies included in our meta-analysis (more details below; see also Figure S1 and Figure S4).

Data extraction

Data from each article were extracted by one of three authors (SD, TR, JMGS), with another screener verifying the procedure, and four authors (SD, TR, JMGS, AST) collectively discussing and revising unclear cases. We prioritised extracting primary data (i.e., means, variances and sample size for each control-treatment comparison) from text and tables, and if that was not possible, from figures (e.g., boxplots) using the R package 'metaDigitise' v1.0.1 [48]. If primary data were not available, we extracted inferential statistics (e.g., t and F values, contingency tables) to transform them into effect sizes (details below). Last, if these steps failed but the article had openly archived data, we calculated the means, standard deviations and sample sizes from each group ourselves ($n = 4$ studies). We extracted the number of nests in each group as the sample size rather than the number of individuals wherever possible (discussed below).

To capture differences in the type of experimental setup used in the studies, we coded each control-treatment comparison into an 'experimental design' moderator, with the following three levels: (i) non-aromatic GNM vs. aromatic GNM, (ii) no material vs. aromatic GNM, and (iii) no material vs. non-aromatic GNM. Where "non-aromatic GNM" refers to nests containing non-aromatic material (e.g., grass, moss, etc.), "aromatic GNM" refers to nests containing aromatic material added by birds (e.g., *Lavandula stoechas*, *Marrubium vulgare*, etc.), researchers, or both; and "no material" refers to nests that had no material added by researchers or where any plant material added by birds was removed. Importantly, our conclusions remained robust to the exclusion of the no material vs. non-aromatic GNM category ($n = 3$ studies; Table S3), so we kept this category in our dataset following our pre-registered plan. A fourth experimental design compared enhanced supplementation of GNM by the researchers (i.e., researchers + birds) to GNM added by birds alone (n = 2 studies). This experimental design represents a fundamentally different comparison focusing on potential additive effects of GNM, comparing different amounts of aromatic GNM. Therefore, we did not include this fourth category in the meta-regression testing the ‘experimental design’ moderator. In studies where three groups were compared (n = 5), we extracted all possible pairwise comparisons, and accounted for (shared group) non-independence in our analyses (more below).

For each study, we also extracted information related to the study subjects (bird species, geographical location of the study population) and the source from which we extracted data (e.g., figures, raw data files). We categorised the reported fitness proxies into 6 broad categories: physiology, morphology, reproduction, behaviour, parasite and pathogenic load, and phenology. For studies testing whether GNM affects the number of parasites or pathogens found in offspring and/or nests, we categorised them into arthropods (e.g., fleas) or microorganisms (e.g., bacteria). For a complete list of variables extracted, see Figure S7. During data extraction, we encountered some traits (e.g., Leukocyte measures, parent mass; k = 15 effect sizes, n = 5 studies) that we did not include in any of our analyses because it was unclear from the study and our own judgement how these traits related to fitness (see Table S2 for the reasons of exclusion).

We contacted four corresponding authors to obtain missing data (e.g., non-reported results, missing sample sizes). All four authors responded to our email, but only two provided the data as requested; one did not have the data anymore, and one did not give a conclusive response. Additionally, to locate grey literature (e.g., unpublished data), we used a standardised email template (Figure S5) to write to the corresponding author(s) of all the 23 articles that passed full-text screening asking for any potential unpublished data that could be included in our meta-analysis. We obtained unpublished data for a study on blue tits and a study on common buzzards (*Buteo buteo*; see description in Figure S6).

Effect size calculation

Using the means, standard deviations, and sample sizes for each control-treatment comparison, we calculated two effect sizes: a log response ratio (lnRR) and a standardised mean difference with heteroskedasticity correction (SMDH), using the ‘escalc’ function (measure = ‘ROM’ and ‘SMDH’, respectively, vtype = ‘LS’) in the R package ‘metafor’ v4.8-0 [49]. We specified the treatment group as the numerator and the control group as the denominator for lnRR, and treatment minus control for SMDH, such that positive values indicate a positive effect of GNM on fitness. In cases such as measures of parasite load or prevalence, laying or hatching dates, courtship time, or scab scores, the effect size sign was inverted before analysis to ensure that a positive value consistently meant an increase in fitness across all fitness proxies (i.e., to ensure comparability across effect sizes). Since SMDH can only be calculated when the degrees of freedom (i.e., $n1+n2-2$) are greater than 1, we had to exclude nine effect sizes from one study that did not fulfil that requirement.

When we could not extract means and variances but inferential statistics such as t and F values (k = 4, n = 2), we transformed them into SMD values [50]. For dichotomous variables (e.g., nest success, recorded as successful vs unsuccessful and analysed as counts per group, k = 4, n = 2), we calculated log odds ratios (lnOR) using the 'escalc' function (measure = OR) from 'metafor' and subsequently converted them to SMD [50]. Since lnRR cannot be calculated from inferential statistics, dichotomous variables, or traits including zero or negative values (k = 7), the number of effect sizes included in the lnRR subset (k = 259, n = 28) is smaller than that of SMDH (k = 274, n = 28).

When calculating each effect size's sampling variance, we used the number of nests per group, rather than the number of individuals, whenever possible, to account for within-nest non-independence due to shared genetic background and early-life environment. In cases where the same control or treatment group was used in multiple comparisons, sample sizes were adjusted before calculating effect sizes by dividing them by the number of times each group was used in a comparison to account for shared group non-independence. For cases where the authors of an original article reported a statistically non-significant difference between two groups but without providing numerical values (k = 16, n = 2), we assigned zero as the effect size for that comparison and imputed its sampling variance using the missing-case approach described in [51] for lnRR, and using $\frac{n_1+n_2}{n_1 \times n_2}$ for SMDH [52].

Data analysis

We performed all analyses in R v4.5.1 [53]. All multilevel meta-analytical models and meta-regressions were fitted using 'metafor' v4.8-0 [49]. Sampling variances were modelled as variance-covariance matrices assuming a 0.5 correlation (ρ) between sampling variances from the same study [54] using the 'vcalc' function from 'metafor' (see Supplementary Appendix Figure S9 for sensitivity analyses showing that conclusions remained the same when using different values of ρ).

We originally considered the following levels of non-independence for our analyses: multiple effect sizes collected from (a) same study (Paper ID), (b) bird species (Species ID), (c) geographical locations (Population ID), (d) sub-populations within the same location (Experiment ID; e.g., birds studied at the same location for multiple years), (e) groups of individuals within a population (Group ID; e.g., males and females in a population, first clutch and second clutch), and (f) repeated measurement of the same trait for the same individuals (Repeated trait ID; e.g., chick mass on day 7, 14 and 21). In the end, our random effect structure deviated from our pre-registration because Experiment ID and Group ID represented nearly the same grouping structure, so we left out the one with lesser resolution (i.e., Experiment ID). Additionally, few effect sizes were measured repeatedly, making Repeated trait ID almost identical to Observation ID (Figure S8), so we excluded Repeated trait ID. Last, we added Population ID since we noticed after extracting the data, but before any analyses, that several studies were performed in the same geographical locations. Unless stated otherwise, we fitted the following random effects in all our models to account for non-independence: Paper ID, Group ID, Species ID, Population ID, and an observation-level random effect (Observation ID).

Meta-analytic mean and hypotheses: To estimate the effect of GNM on fitness estimates across species, we fitted two separate sets of models for each effect size: lnRR and SMD(H). First, we fitted multilevel intercept-only meta-analytical models to test the overall effect of GNM on fitness estimates, its uncertainty and heterogeneity. We used the 'rma.mv' function from the 'metafor' package (method = 'REML' and test = 't' for calculating the test statistics and confidence intervals for the moderators). Although we excluded traits for which their link with fitness is debated in the literature from our main analysis (e.g., sex ratio [55]; yolk hormones levels, [56]; more in Table S2), we conducted a sensitivity analysis including these traits that confirmed the robustness of our results (Table S3; k = 21, n = 8). For all intercept-only models, we estimated the total amount of absolute heterogeneity (σ^2; i.e., the total variation not explained by sampling error), its sources or relative heterogeneity (I^2), mean-standardised metric (CVH2) and variance-mean-standardised metric of heterogeneity (M2) following the pluralistic approach suggested by [57].

We ran several additional sensitivity analyses to test the robustness of the overall results to the: (i) use of pre-registered bivariate multilevel meta-analytic model including both effect sizes as response variables; (ii) inclusion of traits for which their relationship with fitness was debated in the literature (non-pre-registered); (iii) removal of effect sizes that were originally reported as simply statistically non-significant and we assumed them to be zero (non-pre-registered); (iv) both (ii) and (iii) (non-pre-registered); (v) removal of effect sizes from non-aromatic vs. aromatic experimental design (non-pre-registered); (vi) removal of effect sizes from no material vs. non-aromatic experimental design (non-pre-registered); (vii) removal of effect sizes that were calculated by transforming inferential statistics (pre-registered); (viii) removal of effect sizes that failed the Geary's Test for lnRR (non-pre-registered); (ix) removal of unpublished data (non-pre-registered); and (x) use of different values of correlation (ρ) between sampling variance for variance-covariance matrix (pre-registered). The number of effect sizes (k) included in those sensitivity analyses ranged from 131 to 274 (Table S3). In all cases, the results were qualitatively unchanged, indicating that our main conclusions were robust to these alternative criteria.

To test the relative importance of the two main functional hypotheses suggested for GNM, that is, the courtship and the parental care hypothesis (which combines the nest protection and drug hypotheses), we ran a multilevel meta-regression including the moderator 'hypothesis' (levels: courtship, parental care, both, where 'both' refers to fitness proxies not uniquely assigned to either mechanism because they may reflect both courtship and parental care, e.g., offspring morphology or survival). For all meta-regressions, we estimated $R^2_{marginal}$, which corresponds to the percentage of heterogeneity explained by the moderators [58].

Exploratory meta-regressions: We performed additional pre-registered multilevel meta-regressions for which we did not have clear directional predictions. To understand the effect of the type of experimental design (i.e., whether the authors compared non-aromatic vs. aromatic, no material vs. aromatic, no material vs. non-aromatic), we ran a meta-regression with the type of 'experimental design' as the moderator. To test the effect of time of addition of GNM on fitness (levels: before egg hatching, after egg hatching, continuously throughout the nesting phase), we ran a meta-regression with 'time of addition of GNM' as the moderator. We also ran a meta-regression using 'bird species' as the moderator (levels: *Buteo buteo*, *Cyanistes caeruleus*, *Passer cinnamomeus*, *Parus major*, *Sturnus unicolor*, *Sturnus vulgaris*, *Tachycineta bicolor*) to understand if and how the effect of GNM on fitness might differ among species. Furthermore, we explored whether the effect of GNM on fitness varied across the different 'category of fitness proxy' studied (levels: behaviour, morphology, parasite and pathogenic load, phenology, physiology, reproduction) by running a meta-regression with this moderator. Last, for the data subset on parasites and pathogens (k = 90, n = 16), we ran a meta-regression with 'type of parasite' as the moderator (levels: arthropod, micro-organism).

In addition, we performed three additional pre-registered multilevel meta-regressions to explore differences in internal validity across the studies associated with our conclusions (i.e., Risk of Bias, or RoB). Specifically, we evaluated the effects of (i) blinding (whether outcome assessors were blinded to the experimental group), (ii) random assignment to treatment, and (iii) partial reporting (whether all measured outcomes were completely reported or only a subset). For each study, each of these three moderators were coded (levels: yes, no) and included in a separate meta-regression for both lnRR and SMD(H). These meta-regressions aimed at exploring how features associated with risk of bias may explain additional heterogeneity among effect sizes [59,60].

Publication Bias Test: Publication bias occurs when a subset of research findings, most often statistically non-significant ones, are less likely to be published or published later [61]. Two types of publication biases are commonplace in ecology and evolutionary biology: small-study (i.e., small sample size studies reporting larger overall effect sizes due to missing effect sizes) and decline effects (i.e., effect sizes decreasing over time) [62]. We tested them for both lnRR and SMD(H) separately, following [63], first for published studies only (i.e., excluding the two unpublished datasets), then for all studies. To test for small-study effects, we ran a multilevel meta-regression using the 'square-root of the inverse of the effective sample size' as the moderator (i.e., $\sqrt{(n_1 + n_2)/(4n_1n_2)}$). Note that the intercept of this meta-regression provides a sensivity test of the overall estimate adjusted for small-study effects. To test for decline effects (also known as time-lag bias, [64]), we ran a multilevel meta-regression using 'year of publication' (mean-centered) as the moderator. Throughout, 95% CI and 95% PI refer to 95% confidence intervals and 95% prediction intervals, respectively.

Results

We obtained 274 effect sizes from 28 studies conducted on 7 species across 17 geographical locations (Figure 1). About 81% of the effect sizes came from only three bird species (*Cyanistes caeruleus*, k = 77; *Sturnus unicolor*, k = 60; *Sturnus vulgaris*, k = 86). Our dataset included studies published between 1988 and 2024. The mean (± standard deviation) number of effect sizes calculated per study was 9.8 ± 6.4 (median: 8.5, range: 2–24) and the mean effective sample size (i.e., the number of nests/individuals per effect size after accounting for shared-group non-independence) was 32.3 ±31.1 (median: 22.0, range: 3–210 nests/individuals; 31 data points had a sample size of less than 5 nests in each group).

Meta-analytic mean

Both lnRR and SMD(H) intercept-only meta-analyses showed an average positive effect of GNM on the fitness proxies; however, this overall effect was only statistically significant for SMD(H). For lnRR, nests with an experimental increase in GNM showed an average statistically non-significant increase in fitness estimates of 1.9% (Figure 2 A; lnRR = 0.019, 95% CI: [-0.012, 0.050], 95% PI: [-0.281, 0.319], p-value = 0.223, k = 238, n = 26) but the heterogeneity among effect sizes was high. Total raw heterogeneity (σ^2) was 0.023 (Q = 2277.7, p < 0.001), the measures of relative (I^2_{total} = 98.6%) and standardised heterogeneity ($CVH2_{total}$ = 63.59; $M2_{total}$ = 0.98) all suggested high heterogeneity across effect sizes, exceeding the 75th percentile of empirically derived estimates from meta-analyses in ecology and evolution [57].

For SMD(H), nests with an experimental increase in GNM showed an average statistically significant increase in fitness estimates (Figure 2 B; SMD(H) = 0.179, 95% CI: [0.048, 0.310], 95% PI: [-0.829, 1.188], p-value = 0.008, k = 253, n = 26), but the heterogeneity among effect sizes was moderate to high. Total raw heterogeneity (σ^2) was 0.258 (Q = 1114.7, p < 0.001), and both relative (I^2_{total} = 66.1%) and standardised heterogeneity metrics ($CVH2_{total}$ = 8.02; $M2_{total}$ = 0.89) lay between the 50th and 75th percentile of empirically derived estimates [57].

Functional hypotheses

Our multilevel meta-regressions found no clear support for either of the two hypotheses separately (Figure 3). The estimates for the courtship hypothesis ($lnRR_{courtship}$ = 0.039, 95% CI: [-0.172, 0.250], 95% PI: [-0.329, 0.407], p-value = 0.716, k = 8, n = 2; $SMD(H)_{courtship}$ = 0.171, 95% CI: [-0.360, 0.702], 95% PI: [-0.964, 1.305], p-value = 0.527, k = 8, n = 2), and the “parental care hypothesis” ($lnRR_{parental\_care}$ = -0.010, 95% CI: [-0.065, 0.045], 95% PI: [-0.316, 0.296], p-value = 0.716, k = 106, n = 19; $SMD(H)_{parental\_care}$ = 0.109, 95% CI: [-0.060, 0.278], 95% PI: [-0.908, 1.126], p-value = 0.204, k = 115, n = 19) were statistically non-significant. However, the third level in this meta-regression (“Both”), which contained about half of all effect sizes that could not be unequivocally assigned to only one hypothesis, showed an average positive GNM effect on the fitness estimates, statistically significantly so for SMD(H) ($lnRR_{both}$ = 0.028, 95% CI: [-0.006, 0.062], 95% PI: [-0.275, 0.331], p-value = 0.110, k = 124, n = 22; $SMD(H)_{both}$ = 0.231, 95% CI: [0.079, 0.382], 95% PI: [-0.784, 1.245], p-value = 0.003, k = 130, n = 23). None of the three categories differed significantly from each other ($p\text{-value}_{lnRR}$ > 0.211 and $p\text{-value}_{SMD(H)}$ > 0.178 in all cases), and the moderator explained little heterogeneity ($\mathrm{R}^2_{\mathrm{marginal(lnRR)}}$= 1.6% and $\mathrm{R}^2_{\mathrm{marginal(SMD(H))}}$= 1.4%).

Exploratory meta-regressions

First, the overall effect of GNM on fitness estimates differed largely depending on the experimental design used in each study (Figure 4 A,B). The overall effect size for the comparison between no material and aromatic GNM was positive and statistically significant ($lnRR_{no\text{-}material\text{-}vs\text{-}aromatic}$ = 0.111, 95% CI: [0.021, 0.200], 95% PI: [-0.225, 0.446], p-value = 0.016, k = 90, n = 12 and $SMD(H)_{no\text{-}material\text{-}vs\text{-}aromatic}$ = 0.712, 95% CI: [0.309, 1.114], 95% PI: [-0.826, 2.250], p-value = 0.001, k = 94, n = 12], whereas, the comparison between non-aromatic GNM and aromatic GNM was statistically non-significant (Table 1). The comparison between no material and non-aromatic GNM was negative and, although based on only 3 studies, statistically significant for lnRR ($lnRR_{no\text{-}material\text{-}vs\text{-}non\text{-}aromatic}$ = -0.349, 95% CI: [-0.504, -0.195], 95% PI: [-0.708, 0.009], p-value <0.001, k = 10, n = 3; Table 1). All levels differed statistically significantly from each other except one ($p\text{-}value_{lnRR}$ < 0.004 and $p\text{-}value_{SMD(H)}$ < 0.001 in all cases, except $SMD(H)_{no\text{-}material\text{-}vs\text{-}aromatic}$ vs. $SMD(H)_{no\text{-}material\text{-}vs\text{-}non\text{-}aromatic}$ p-value = 0.140), and the moderator experimental design explained a large proportion of heterogeneity ($R^2_{marginal(lnRR)}$= 27.8% and $R^2_{marginal(SMD(H))}$= 24.4%).

Second, the overall positive effect of GNM on fitness outcomes was largest when material was added continuously throughout the nesting phase (Figure 4 C,D). For continuous addition, the overall effect was positive for lnRR and SMD(H) but statistically significant only for SMD(H) ($lnRR_{continuously}$ = 0.034, 95% CI: [-0.014, 0.083], 95% PI: [-0.270, 0.338], p-value = 0.165, k = 95, n = 12; $SMD(H)_{continuously}$ = 0.302, 95% CI: [0.069, 0.534], 95% PI: [-0.726, 1.329], p-value = 0.011, k = 105, n = 12), whereas the effect was small and statistically non-significant when added before egg hatching (Table 1). Last, the overall effect was positive for SMD(H) when GNM was added after egg hatching only, although the confidence interval overlapped zero ($SMD(H)_{after\_egg\_hatching}$ = 0.267, 95% CI: [-0.024, 0.559], 95% PI: [-0.775, 1.310], p-value = 0.072, k = 57, n = 5; Table 1). The three categories did not differ statistically significantly from each other ($p\text{-}value_{lnRR}$ > 0.307 and $p\text{-}value_{SMD(H)}$ > 0.243 in all cases). The timing of GNM addition explained little heterogeneity ($R^2_{marginal(lnRR)}$= 1.2% and $R^2_{marginal(SMD(H))}$= 4.5 %).

Third, although none of the species-specific mean estimates were statistically significant, most were positive (Figure S13, Table 1), and bird species explained some heterogeneity ($R^2_{marginal(lnRR)}$= 5.3% and $R^2_{marginal(SMD(H))}$ = 8.5%). Fourth, most fitness proxy categories had positive mean estimates, but only morphology was statistically significant, and only for SMD(H) ($SMDH_{morphology}$ = 0.310, 95% CI: [0.112, 0.509], 95% PI: [-0.723, 1.343], p-value = 0.002, k = 65, n = 15; Figure S13; Table 1) and fitness proxy category accounted for a small amount of heterogeneity ($R^2_{marginal(lnRR)}$= 3.7% and $R^2_{marginal(SMD(H))}$= 2.4%). Fifth, for the subset of outcomes relating to pathogen and parasitic load, we found no statistically significant difference between arthropods and micro-organisms (Figure S13 ; Table 1) and the type of parasite or pathogen explained little heterogeneity ($R^2_{marginal(lnRR)}$= 0.4% and $R^2_{marginal(SMD(H))}$= 1.6%). Lastly, the evaluation of potential risk of bias categories (i.e., blinding, random assignment to each treatment, and partial reporting) was limited by the small number of studies in some categories (details in Table S4)

Publication Bias

We found no conclusive evidence for the existence of publication bias in the published literature (Figure S14). There was no statistical evidence for small-study effect for lnRR ($lnRR_{intercept}$= 0.003, 95% CI: [-0.066, 0.072], p-value = 0.926, $lnRR_{slope}$ = 0.079, 95% CI: [-0.215; 0.373], p-value = 0.595, k = 221, n = 24, $R^2_{marginal(lnRR)}$ = 0.3%). For SMD(H), effect sizes were larger when sample size was smaller, but the confidence interval overlapped zero ($SMDH_{slope}$ = 1.393, 95% CI: [-0.143, 2.929], p-value = 0.075, k = 235, n = 24, $R^2_{marginal(SMD(H))}$= 8.3%), a pattern that was ameliorated by including the two unpublished studies ($SMDH_{slope}$ = 1.218, 95% CI: [-0.225, 2.661], p-value = 0.098, k = 253, n = 26, $R^2_{marginal(SMD(H))}$= 6.7%). The overall effect adjusted for small-study effects was not statistically different from zero ($SMDH_{intercept}$ = -0.035, 95% CI: [-0.319, 0.249], p-value = 0.809). In addition, since published effect sizes did not change over time, our analyses suggest no clear evidence of decline effects in published effect sizes (see Figure S14).

Discussion

Our meta-analysis of experimental studies shows that green nest material (GNM) has a positive effect on fitness proxies across the bird species studied. Our results were generally consistent across two effect size metrics and robust to several sensitivity analyses. However, moderate-to-high heterogeneity among effect sizes indicates that the benefits of GNM are context-dependent. We found no clear support for either of the two functional hypotheses traditionally proposed separately, that is, the courtship hypothesis and the parental care hypothesis (i.e., nest protection and drug hypotheses combined). While several studies investigated the courtship hypothesis, only a few fitness proxies could be exclusively linked to this mechanism (e.g., courtship time, male provisioning), and these showed no clear effect of GNM. Similarly, fitness proxies exclusively associated with the parental care hypothesis (e.g., parasite loads, nestling health), also showed no clear effect of GNM. This finding is particularly surprising because the parental care hypothesis has been most frequently tested. While studies testing the courtship hypothesis have suggested that the aromatic volatile compounds could help receivers of the signal detect the presence of GNM in dark cavity nests [26], it is the parental care hypothesis that has stressed the importance of these compounds [65–67]. Our findings challenge this assumption since experimental designs comparing aromatic with non-aromatic GNM, which account for more than half of all tests, did not show a clear fitness effect, whereas comparisons between aromatic GNM and no material did. By contrast, comparisons between non-aromatic GNM and no material suggested a negative effect of non-aromatic GNM on fitness, although this result should be interpreted cautiously because it was based on only three studies (k = 10 effect sizes). Notably, experimental design alone explained about one-quarter of the observed heterogeneity.

One possible explanation is that comparing aromatic vs. non-aromatic isolates the effects of aromaticity or volatile compounds, which may be weak, context-dependent (e.g., dependent on parasite pressure, [14]), or short-lived due to rapid decay of volatiles [15] and may require repeated replenishment [68]. By contrast, comparing aromatic vs. no material captures the net effect of adding green material itself, including potential non-chemical and volatile effects, which may therefore show a clearer overall benefit. Such benefits could also relate to plant morphology such as shape or colour. In addition, only proxies that could not be clearly assigned to either of the functional hypotheses (e.g., survival rate, nestling morphology) and were therefore categorised as being associated with "both" hypotheses, showed a clear positive effect. This ambiguity reflects the difficulty of linking many fitness proxies to an underlying function since they can be shaped by multiple overlapping processes. For example, nestling mass, a commonly measured fitness proxy, could still be influenced by pathogen load or direct physiological effects of plant compounds, even though our synthesis found no consistent overall evidence for parasite reduction or health benefits. Nestling mass, however, could also respond to altered parental investment through differential allocation, making such broad proxies of fitness difficult to assign to a single function.

Heterogeneity in the overall effect of GNM was mainly associated with within-study variation, with little to no among-study variation. This suggests that, once methodological (e.g., experimental design) and biological factors (e.g., species identity, category of fitness proxy) are accounted for, the overall effect found would be potentially generalisable across studies. In addition to experimental design, continuous supplementation of GNM throughout the nesting period had stronger positive effects than supplementation restricted to before or after hatching. Although not conclusive, this pattern is more consistent with the parental care hypothesis than with the courtship hypothesis. In blue tits, the concentration of active compounds released by the aromatic plants can decrease within 24 hours [15], and females often replace experimentally removed herbs [39,68]. This repeated addition may help maintain an environment rich in volatiles that is hostile to parasites; however, fresh GNM could also improve fitness through structural or microclimate benefits [2,42]. Regular replenishment of GNM by birds may help nest lining as material desiccates, thereby improving insulation and humidity buffering [42,69] and diluting faecal matter or food debris in the nest (i.e., hygiene function, [70]). This microclimate stability may accelerate nestling development while reducing thermoregulatory costs for the parents [71,72].

Among the biological moderators tested, species identity was the most important one, providing some evidence for species-specific effects; however, three species were represented in only one study each. *Tachycineta bicolor* is a notable exception among the included species because it does not naturally incorporate GNM. Experiments in this species can help disentangle direct aromatic effects from courtship-related functions, but the available data showed no clear fitness benefit of aromatic GNM addition in this species. Most included species were cavity-nesting passerines studied in nest boxes; broad life-history differences are unlikely to fully explain species-specific variation. Instead, such variation may reflect differences in GNM-use that may arise from how, when, and by which sex GNM is added, whether species naturally use GNM, and the ecological contexts in which nests occur, including parasite pressure and their interplay.

The second most important biological moderator was the fitness proxy category, potentially indicating differences in validity across fitness proxies or reflecting differences among underlying mechanisms. Morphological outcomes (e.g., nestling mass, tarsus length) showed the largest positive effects, suggesting that GNM has a more pronounced effect on offspring development or growth. Morphology was the second most commonly measured fitness proxy across studies, likely because morphological traits can be obtained easily and non-invasively in the field as compared to long-term demographic (e.g., survival, recruitment) or physiological assays (e.g., blood-based immune or endocrine measures). However, while widely used as an indicator of offspring condition, these morphological traits, like many other outcomes in the literature, are indirect fitness proxies, and their relationship to recruitment or survival may be context dependent [73]. The positive effect of GNM on morphological traits may also reflect that morphology responds quickly to within-nest conditions, whereas effects on long-term survival may be diluted by subsequent environmental variations [74].

All other fitness proxy categories, including physiology, reproduction, phenology, and behaviour, showed small to moderate positive effects, whereas proxies related to parasite or pathogen load, despite being the most commonly investigated, showed weak and inconsistent effects. Parasite reduction is central to one of the two mechanisms within the parental care hypothesis (the nest protection hypothesis, [14,30]). Our results further emphasise that despite its prominence, there is a lack of clear overall evidence for the nest protection hypothesis. The other hypothesis categorised within the parental care hypothesis (i.e., the drug hypothesis, [19,35]), proposes an increase in offspring survival, with volatile compounds directly enhancing offspring immune function and/or health independent of parasite or pathogen control. While the largest positive effects of morphology could support the drug hypothesis, it should be noted that (i) the mechanism linking volatiles to enhanced offspring immune function or health is unclear, (ii) our results suggest that non-aromatic GNM may play a role, and (iii) several other mechanisms could also be at play such as nest microclimate and incubation conditions [75] or parental investment.

When interpreting our results, it is important to acknowledge some constraints of our study and the available evidence. The overall effect of GNM differed between the two effect size metrics, which make distinct assumptions regarding the underlying biological processes and the statistical properties. Indeed, excluding effect sizes where lnRR failed the Geary's test reduced such differences Table S3. Overall, these divergent results highlight the importance of choosing effect size metrics for a meta-analysis that align with the biological data at hand or, in cases where several different types of measures are present, using both [76]. We found high reporting quality in the primary studies included in our meta-analysis, with 15.4% partial reporting, and only some evidence of small-study effects for SMD(H) analyses. Nonetheless, many studies had very low sample sizes (41% of the effect sizes were based on fewer than 10 nests per group), which reduces power and increases heterogeneity [77]. Furthermore, research in this area remains heavily biased towards just three bird species, limiting our ability to infer broad patterns. Expanding the scope beyond the currently well-studied taxa would be essential to understand generalisability across species. Specifically, targeting species with diverse nesting ecologies and habitat conditions can help us better understand the adaptive benefits of this behaviour.

Conclusion

After more than 40 years of research on this topic, ours is the first quantitative synthesis confirming an overall effect of green nest material on fitness. Despite moderate-to-high heterogeneity, our results suggest that the effect may be generalisable across studies once methodological and biological factors within studies are accounted for. Our small-study effects analyses, although non-conclusively, indicate that some studies may remain unpublished, despite our efforts to include unpublished work. Moreover, our study emphasises that the mechanism remains largely unclear and challenges the assumption that volatile compounds alone explain the relationship between green nest material and fitness. Future experimental designs should focus on decoupling volatiles from structural or microclimate effects, while quantifying mediators (e.g., nest microclimate, parasite load) and prioritising fitness proxies more directly linked to fitness (e.g., fledging success, recruitment), as well as testing explicitly, rather than assuming, the role of these volatiles. Replications across parasite and climatic gradients, together with cross-fostering experiments, could allow us to understand the context dependency and distinguish nest-level effects from parental quality. In addition, theoretical work could further solidify the field’s foundations and yield testable predictions. Cost-benefit models of GNM could help predict when net benefits are expected to occur and why effects vary among species and environments, generating concrete predictions (e.g., benefits arising only above certain parasite pressure threshold or under certain temperature or humidity conditions) that are directly testable with multi-site experiments. Altogether, our synthesis contributes to a more nuanced understanding of the adaptive significance of one of the most fascinating and most studied natural structures—bird nests, opening doors to several questions that remain elusive.

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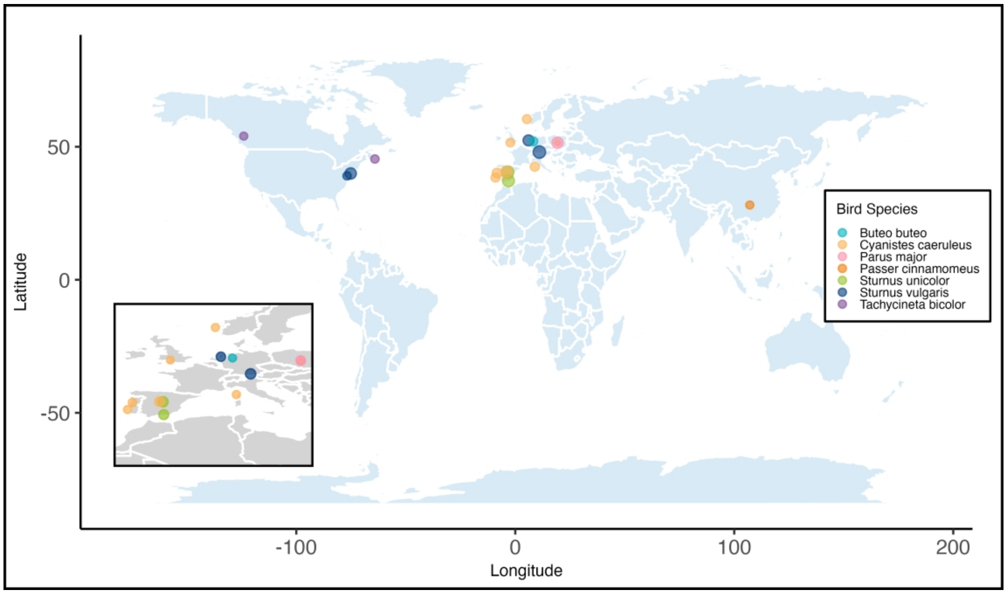

Geographical distribution of the 17 study populations included in the meta-analysis testing the adaptive value of green nest material in birds. Point size reflects the number of effect sizes calculated from each location (range: 3–34), and colour denotes species (see legend), with an inset zooming in on a dense cluster of data points.

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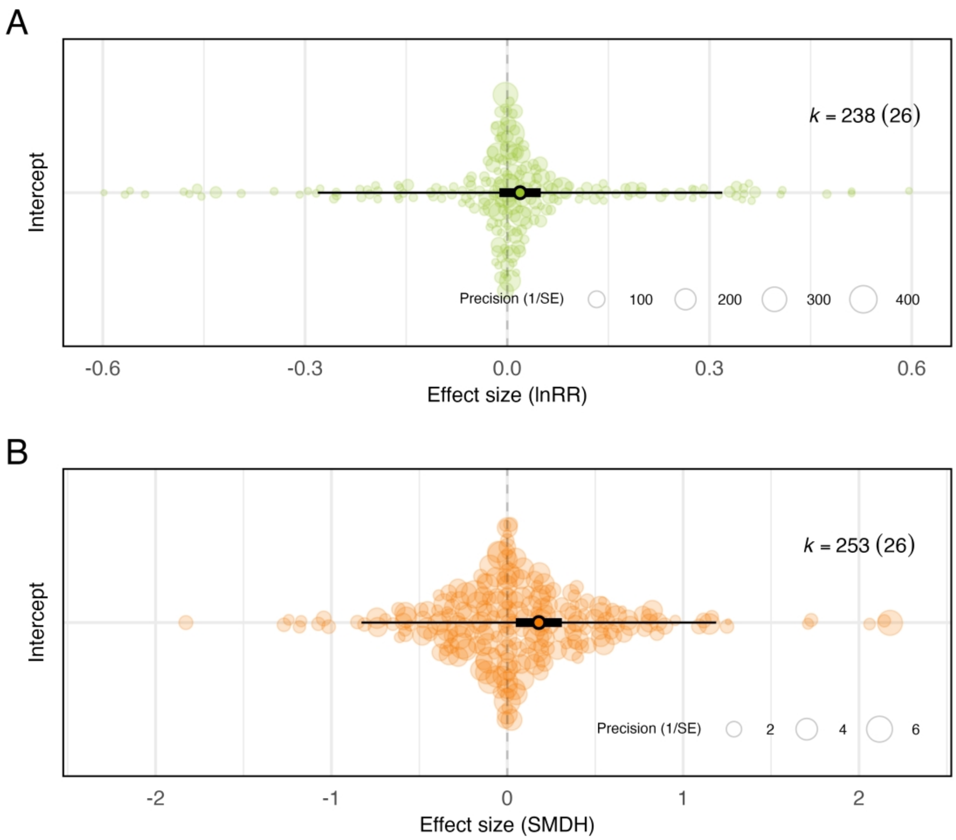

Overall effect of green nest material on all fitness proxies combined. The meta-analytic mean (black-border circle), 95% confidence intervals (thicker bars) and prediction intervals (thinner bars) are shown along with the data points (coloured circles) scaled by precision (inverse of standard error). Results are shown as an orchard plot for the intercept-only model (A) using log response ratio (lnRR), and (B) standardised mean difference (SMD(H)). k: number of effect sizes (number of studies is shown in brackets). X-axes are truncated for legibility, which partially explains the apparent asymmetry (see Figure S10).

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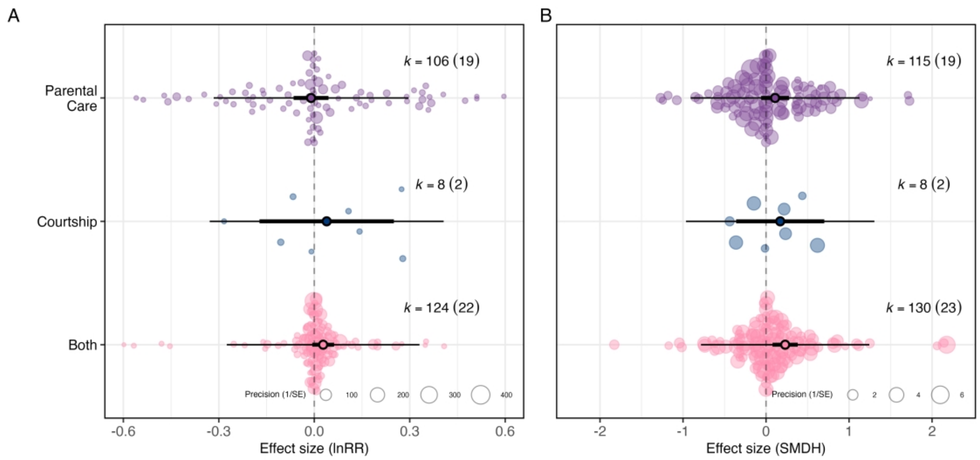

There is no clear evidence for the courtship or the parental care hypothesis separately as drivers of green nest material fitness effects. Orchard plot of the multilevel meta-regression for (A) log response ratio (lnRR) and (B) standardised mean difference (SMD(H)). Mean estimates (black-border circles), 95% confidence intervals (thicker bars), and prediction intervals (thinner bars) are shown along with the individual data points (transparently coloured circles) scaled by precision (inverse of standard error). k: number of effect sizes (number of studies is shown in brackets). X-axes are truncated for legibility, which partially explains the apparent asymmetry (see Figure S11).

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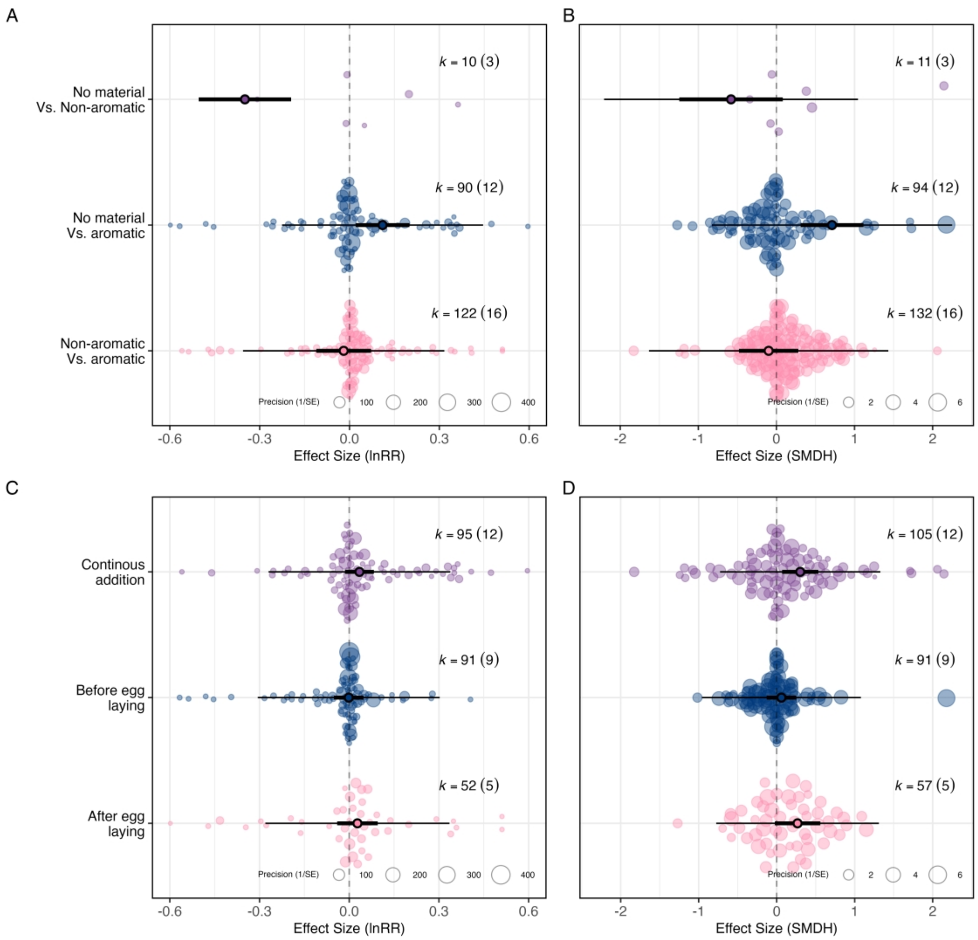

Experimental designs comparing nests with aromatic green nest material against nests with no green material, and those in which green nest material was experimentally added continuously throughout the nesting phase, lead to the largest fitness estimates. Orchard plots of the multilevel meta-regressions showing the overall effect of green nest material depending on experimental design (A: log response ratio (lnRR); B: standardised mean difference (SMD(H)), and time of green nest material addition (C: lnRR; D: SMD(H)). Mean estimates (black-border circles), 95% confidence intervals (thicker bars) and prediction intervals (thinner bars) are shown along the individual data points (transparently coloured circles) scaled by precision (inverse of standard error). k: number of effect sizes (number of studies is shown in brackets). X-axes are truncated for legibility, which partially explains the apparent asymmetry (see Figure S12).

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Multilevel Meta Regressions for Exploratory Analyses

	lnRR						SMDH					
Level	**Estimate**	**P-val**	**95% CI**	**95% PI**	**k**	**n**	**Estimate**	**P-val**	**95% CI**	**95% PI**	**k**	**n**
1. Experimental Design												
Non-aromatic vs. Aromatic	-0.019	0.677	[-0.111, 0.072]	[-0.356, 0.317]	122	16	-0.100	0.606	[-0.479, 0.280]	[-1.632, 1.432]	132	16
No added material vs. Aromatic	0.111	**0.016**	[0.021, 0.200]	[-0.225, 0.446]	90	12	0.712	**0.001**	[0.309, 1.114]	[-0.826, 2.250]	94	12
No added material vs. Non-aromatic	-0.349	**<0.001**	[-0.504, -0.195]	[-0.708, 0.009]	10	3	-0.581	0.085	[-1.244, 0.081]	[-2.207, 1.044]	11	3
2. Time of GNM Addition												
After egg hatching	0.028	0.425	[-0.040, 0.095]	[-0.280, 0.335]	52	5	0.267	0.072	[-0.024, 0.559]	[-0.775, 1.310]	57	5
Before egg hatching	-0.002	0.948	[-0.051, 0.048]	[-0.306, 0.303]	91	9	0.061	0.531	[-0.130, 0.251]	[-0.958, 1.080]	91	9
Continuously through nesting phase	0.034	0.165	[-0.014, 0.083]	[-0.270, 0.338]	95	12	0.302	**0.011**	[0.069, 0.534]	[-0.726, 1.329]	105	12
3. Bird Species												
Buteo buteo	-0.109	0.724	[-0.714, 0.497]	[-0.844, 0.626]	9	1	-0.251	0.732	[-1.691, 1.189]	[-2.263, 1.761]	9	1
Cyanistes caeruleus	0.006	0.969	[-0.288, 0.299]	[-0.504, 0.515]	68	10	0.087	0.866	[-0.926, 1.100]	[-1.646, 1.819]	74	10
Parus major	-0.037	0.815	[-0.347, 0.273]	[-0.556, 0.482]	19	1	0.651	0.275	[-0.521, 1.822]	[-1.178, 2.480]	22	1
Passer cinnamomeus	0.228	0.192	[-0.115, 0.570]	[-0.312, 0.767]	6	1	0.955	0.193	[-0.485, 2.396]	[-1.057, 2.968]	6	1
Sturnus unicolor	-0.011	0.941	[-0.308, 0.286]	[-0.523, 0.500]	44	5	0.069	0.896	[-0.967, 1.105]	[-1.676, 1.815]	45	5
Sturnus vulgaris	0.039	0.794	[-0.255, 0.333]	[-0.471, 0.549]	83	6	0.279	0.590	[-0.742, 1.300]	[-1.457, 2.016]	84	6
Tachycineta bicolor	0.092	0.582	[-0.237, 0.422]	[-0.439, 0.623]	9	2	0.204	0.716	[-0.899, 1.308]	[-1.582, 1.991]	13	2
4. Trait Category												
Behaviour	0.034	0.755	[-0.179, 0.247]	[-0.339, 0.406]	8	2	0.153	0.574	[-0.382, 0.688]	[-0.993, 1.299]	8	2
Morphology	0.036	0.109	[-0.008, 0.080]	[-0.273, 0.345]	62	15	0.310	**0.002**	[0.112, 0.509]	[-0.723, 1.343]	65	15
Parasitic and Pathogen Related	-0.036	0.328	[-0.107, 0.036]	[-0.350, 0.278]	86	16	0.112	0.242	[-0.076, 0.299]	[-0.919, 1.143]	90	16
Phenology	0.004	0.965	[-0.156, 0.163]	[-0.341, 0.348]	4	3	0.081	0.779	[-0.484, 0.645]	[-1.080, 1.241]	4	3
Physiology	0.026	0.521	[-0.053, 0.105]	[-0.290, 0.342]	22	9	0.157	0.288	[-0.133, 0.448]	[-0.897, 1.212]	23	9
Reproduction	0.010	0.727	[-0.045, 0.065]	[-0.301, 0.320]	56	16	0.153	0.109	[-0.034, 0.340]	[-0.878, 1.184]	63	19
5. Parasite Type												
Arthropod	0.013	0.898	[-0.186, 0.212]	[-0.780, 0.806]	56	13	0.144	0.292	[-0.126, 0.414]	[-1.107, 1.395]	59	13
Micro-organism	-0.041	0.720	[-0.268, 0.186]	[-0.842, 0.760]	26	6	-0.025	0.890	[-0.385, 0.335]	[-1.299, 1.248]	27	6

Results from multilevel meta-regressions testing how several factors moderate the effect of green nest material on fitness estimates across birds. Columns show model estimates, p-values (P-val), 95% confidence intervals (CI), 95% prediction intervals (PI), number of effect sizes (k), and number of studies (n) for both effect size metrics: log response ratio (lnRR) and standardized mean difference (SMD(H)). Statistically significant results (p-values < 0.05) are shown in bold.

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